

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Project Name: AI FAQ Assistant API

1. Project Details

Field	Details
Team ID	SWTID-2026-5919
Project Name	AI FAQ Assistant API
Phase	Project Design Phase-II
Document	Solution Requirements (Functional & Non-functional)

2. Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through a form
		Validate user registration details
		Store user details securely in MongoDB
FR-2	User Authentication	Login using registered credentials
		Password hashing and verification using bcrypt
		Generate and validate JWT authentication tokens
FR-3	FAQ Management	Create new FAQs
		View and retrieve existing FAQs
		Update FAQ questions and answers
		Delete outdated FAQs
FR-4	FAQ Search and Retrieval	Search FAQs using keywords
		Retrieve FAQs by category
		Return matching FAQ details through REST API
FR-5	AI-Powered Answer Generation	Accept user questions through the API
		Send questions to Google Gemini AI
		Receive and return AI-generated answers
FR-6	Category Management	Create and manage FAQ categories
		Organize FAQs under relevant categories
		Retrieve FAQs associated with a category

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-7	API Response Management	Return responses in JSON format
		Handle invalid requests and missing data
		Provide appropriate HTTP status codes and error messages

3. Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The API should be easy to use and integrate with web and mobile applications. Requests and responses should follow a clear and consistent JSON structure.
NFR-2	Security	User passwords must be securely hashed using bcrypt. JWT-based authentication should protect authorized endpoints, and sensitive information must not be exposed in API responses.
NFR-3	Reliability	The system should process valid requests consistently and handle errors gracefully. Database or AI service failures should return appropriate error responses without crashing the application.
NFR-4	Performance	FAQ retrieval and keyword searches should return responses efficiently. Database queries and API processing should be optimized to reduce response time.
NFR-5	Availability	The API should remain accessible whenever the application server, database, and required external services are operational. Temporary service failures should be handled gracefully.
NFR-6	Scalability	The system should support an increasing number of users, FAQs, and API requests. Its modular Node.js and Express.js architecture should allow future enhancements and scaling.

4. Summary

The AI FAQ Assistant API is designed to simplify customer support by providing a centralized system for FAQ management, keyword-based search, and AI-powered answer generation. It enables users to access information quickly while allowing authorized users to maintain FAQ content and categories.

The functional requirements define the operations provided by the system, including registration, authentication, FAQ management, search, and AI integration. The non-functional requirements address usability, security, reliability, performance, availability, and scalability to support a secure and maintainable API.

The proposed solution uses Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, and Google Gemini AI to implement these requirements.